

**GIRRAWEE HIGH SCHOOL
MATHEMATICS**

Year 12 Extension 2 Task ~~3~~ 4

Thursday 9th June 2005

- Instructions: a) Write all your answers on your own paper.
b) Show all necessary working.
c) Marks may be deducted for careless or badly arranged work.

Time Allowed: 90 minutes

Question 1 (25 marks)

Marks

Find the following integrals:

- | | |
|--|---|
| (i) $\int \frac{1}{x \log x} dx$ | 3 |
| (ii) $\int \frac{x^2}{1+x} dx$ | 3 |
| (iii) $\int \sin^4 x \cos^3 x dx$ | 4 |
| (iv) $\int_0^{\frac{\pi}{2}} \frac{dx}{1 + \sin x + \cos x}$ | 4 |
| (v) $\int x \tan^{-1} x dx$ | 4 |
| (vi) $\int \frac{3x+1}{x^2+2x+2} dx$ | 3 |
| (vii) $\int \frac{x^2 - x - 21}{(2x-1)(x^2+4)} dx$ | 4 |

Question 2 (13 marks)

- a) (i) If $I_n = \int \tan^n x dx$ for $n \geq 0$ show that $I_n = \frac{1}{n-1} \tan^{n-1} x - I_{n-2}$ for $n \geq 2$ 3
- (ii) Evaluate $\int_0^{\frac{\pi}{4}} \tan^7 x dx$ 3
- b) (i) Prove that $\int_0^a f(x) dx = \int_0^a f(a-x) dx$ 2
- (ii) Consider $f(x) = \frac{1}{1 + \tan x}$ where $0 \leq x \leq \frac{\pi}{2}$ and $f\left(\frac{\pi}{2}\right) = 0$ 2
- Show that $f(x) + f\left(\frac{\pi}{2} - x\right) = 1$
- (iii) Hence evaluate $\int_0^{\frac{\pi}{2}} \frac{1}{1 + \tan x} dx$ 3

Question 3 (22 marks)

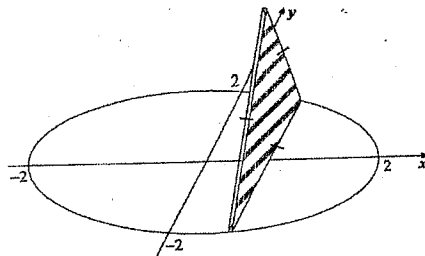
Marks

- a) The region between the curve $y = \sin x$, the line $y = 1$ and the y axis is rotated about the line $y = 1$. Using a slicing technique find the volume formed.

5

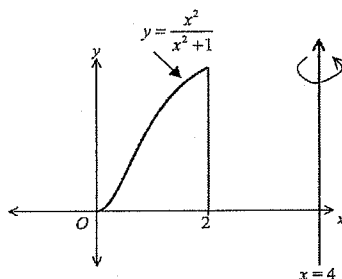
b)

5



The diagram shows a cross-sectional slice of a solid whose base is the region enclosed by the circle $x^2 + y^2 = 4$. Each such cross-section of the solid is an equilateral triangle. Find the volume of the solid.

- c) The region bounded by the curve $y = \frac{x^2}{x^2 + 1}$, the x axis and $0 \leq x \leq 2$, is rotated about the line $x = 4$ to form a solid.



- (i) Using the method of cylindrical shells, explain why the volume ΔV of a typical shell distant x units from the origin and with thickness Δx is given by

3

$$\Delta V = 2\pi(4-x)\left(1 - \frac{1}{1+x^2}\right)\Delta x$$

- (ii) Hence, find the total volume of the solid formed.

3

- d) (i) Show that the area of a regular hexagon of side s is given by $A = \frac{3\sqrt{3}s^2}{2}$

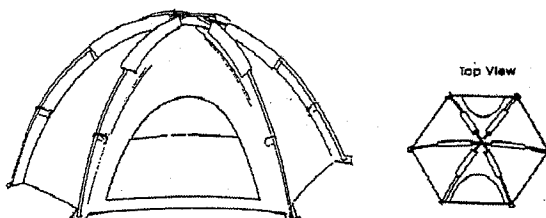
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- (ii) The diagrams below illustrate a dome tent. When erected, the base is a regular hexagon which measures 2 metres from corner to adjacent corner.

4

Flexible exterior poles extend between opposite corners in semi-circle arcs to support the tent.

By taking slices parallel to the base of the tent find the volume enclosed by the tent.



Year 12 Extension 2 Task 3 2005 Solutions

Question 1 (25)

$$\begin{aligned} \text{(i)} \int \frac{1}{x \log x} dx & \quad u = \log x \\ &= \int \frac{du}{u} \\ &= \log u + c \\ &= \log \log x + c \end{aligned} \quad (3)$$

$$\begin{aligned} \text{(ii)} \int \frac{x^2}{1+x} dx & \quad \begin{array}{l} x-1 \quad x-1 \\ x+1 \quad x^2+0x+0 \\ \hline x^2+x \\ -x+0 \\ \hline -x-1 \\ -x-1 \\ \hline 0 \end{array} \\ &= \int \left[x-1 + \frac{1}{1+x} \right] dx \\ &= \frac{1}{2}x^2 - x + \log(1+x) + c \end{aligned} \quad (3)$$

$$\begin{aligned} \text{(iii)} \int \sin^4 x \cos^3 x dx & \quad u = \sin x \\ &= \int \sin^4 x (1 - \sin^2 x) dx \\ &= \int (u^4 - u^6) du \\ &= \frac{1}{5}u^5 - \frac{1}{7}u^7 + c \\ &= \frac{1}{5}\sin^5 x - \frac{1}{7}\sin^7 x + c \end{aligned} \quad (4)$$

$$\begin{aligned} \text{(iv)} \int_0^{\frac{\pi}{2}} \frac{dx}{1 + \sin x + \cos x} & \quad t = \tan \frac{x}{2} \\ &= \int_0^1 \frac{2dt}{1+t^2} \\ &= \int_0^1 \frac{2dt}{1+t^2+1-t^2} \\ &= \int_0^1 \frac{2dt}{2+t^2} \\ &= \int_0^1 \frac{dt}{t^2+1} \\ &= [\log(t+1)]_0^1 \\ &= \log 2 \end{aligned} \quad (4)$$

$$\begin{aligned} \text{(v)} \int x \tan^{-1} x dx & \quad u = \tan^{-1} x \quad v = \frac{1}{2}x^2 \\ &= \frac{1}{2}x^2 \tan^{-1} x - \frac{1}{2} \int \frac{x}{1+x^2} dx \\ &= \frac{1}{2}x^2 \tan^{-1} x - \frac{1}{2} \int \left[1 - \frac{1}{1+x^2} \right] dx \\ &= \frac{1}{2}x^2 \tan^{-1} x - \frac{1}{2}x + \frac{1}{2} \tan^{-1} x + c \end{aligned} \quad (4)$$

$$\begin{aligned} \text{(vi)} \int \frac{3x+1}{x^2+2x+2} dx & \quad \begin{array}{l} 3x+1 \\ x^2+2x+2 \\ \hline 3x+2 \\ -x+0 \\ \hline -x-1 \\ -x-1 \\ \hline 0 \end{array} \\ &= \frac{3}{2} \int \frac{2x+2}{x^2+2x+2} dx - 2 \int \frac{dx}{(x+1)^2+1} \\ &= \frac{3}{2} \log(x^2+2x+2) - 2 \tan^{-1}(x+1) + c \end{aligned} \quad (4)$$

$$\begin{aligned} \text{(vii)} \int \frac{x^2-x-21}{(2x-1)(x^2+4)} dx & \quad \begin{array}{l} A \\ (2x-1) + \frac{Bx+C}{(x^2+4)} = \frac{x^2-x-21}{(2x-1)(x^2+4)} \\ x = \frac{1}{2} \quad \begin{array}{l} 17A = -\frac{65}{4} \\ A = -5 \end{array} \end{array} \\ & \quad \begin{array}{l} x=0 \quad \begin{array}{l} 4A-C = -21 \\ -20-C = -21 \\ -C = -1 \\ C = 1 \end{array} \\ x=1 \quad \begin{array}{l} 5A+B+C = -21 \\ -25+B+1 = -21 \\ B = 3 \end{array} \end{array} \end{aligned}$$

$$\begin{aligned} &= \int \frac{x^2-x-21}{(2x-1)(x^2+4)} dx \\ &= \int \left[\frac{-5}{2x-1} + \frac{3x+1}{x^2+4} \right] dx \\ &= \int \left[\frac{-5}{2x-1} + \frac{3}{2} \cdot \frac{2x}{x^2+4} + \frac{1}{x^2+4} \right] dx \\ &= -\frac{5}{2} \log(2x-1) + \frac{3}{2} \log(x^2+4) + \frac{1}{2} \tan^{-1} \frac{x}{2} + c \end{aligned} \quad (4)$$

Question 2 (13)

$$\begin{aligned} \text{a) } I_n &= \int \tan^n x dx \\ &= \int \tan^{n-2} x \cdot (\sec^2 x - 1) dx \\ &= \int \tan^{n-2} x \sec^2 x dx - \int \tan^{n-2} x dx \\ & \quad u = \tan x \\ & \quad du = \sec^2 x dx \\ &= \int u^{n-2} du - I_{n-2} \\ &= \frac{1}{n-1} u^{n-1} - I_{n-2} \\ &= \frac{1}{n-1} \tan^{n-1} x - I_{n-2} \end{aligned} \quad (3)$$

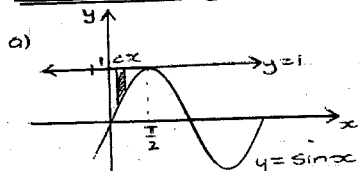
$$\begin{aligned} \text{(ii)} \int_0^{\frac{\pi}{4}} \tan^7 x dx &= \left[\frac{1}{6} \tan^6 x \right]_0^{\frac{\pi}{4}} - I_5 \\ &= \frac{1}{6} - \left[\frac{1}{4} \tan^4 x \right]_0^{\frac{\pi}{4}} + I_3 \\ &= \frac{1}{6} - \frac{1}{4} + \left[\frac{1}{2} \tan^2 x \right]_0^{\frac{\pi}{4}} - I_1 \\ &= \frac{1}{6} - \frac{1}{4} + \frac{1}{2} - \int_0^{\frac{\pi}{4}} \tan x dx \\ &= \frac{5}{12} + [\log \cos x]_0^{\frac{\pi}{4}} \\ &= \frac{5}{12} + \log\left(\frac{1}{\sqrt{2}}\right) \end{aligned} \quad (3)$$

$$\begin{aligned} \text{b) i) } \int_a^b f(x) dx & \quad u = a-x \\ & \quad du = -dx \\ & \quad x=0, u=a \\ & \quad x=a, u=0 \\ &= - \int_a^a f(a-u) du \\ &= \int_a^a f(a-u) du \\ &= \int_0^a f(a-x) dx \end{aligned} \quad (2)$$

$$\begin{aligned} \text{(ii)} f(x) + f\left(\frac{\pi}{2}-x\right) &= \frac{1}{1+\tan x} + \frac{1}{1+\tan\left(\frac{\pi}{2}-x\right)} \\ &= \frac{1}{1+\tan x} + \frac{1}{1+\cot x} \\ &= \frac{1}{1+\tan x} + \frac{1}{1+\frac{1}{\tan x}} \\ &= \frac{1}{1+\tan x} + \frac{\tan x}{\tan x+1} \\ &= \frac{1+\tan x}{1+\tan x} \\ &= 1 \end{aligned} \quad (2)$$

$$\begin{aligned} \text{(iii)} \int_0^{\frac{\pi}{4}} \frac{1}{1+\tan x} dx &= \int_0^{\frac{\pi}{4}} \frac{1}{1+\tan\left(\frac{\pi}{2}-x\right)} dx \\ &= \int_0^{\frac{\pi}{4}} \left[1 - \frac{1}{1+\tan x} \right] dx \\ &= 2 \int_0^{\frac{\pi}{4}} \frac{1}{1+\tan x} dx = \int_0^{\frac{\pi}{4}} dx \\ &= \frac{1}{2} \left[x \right]_0^{\frac{\pi}{4}} \\ &= \frac{\pi}{4} \end{aligned} \quad (3)$$

Question 3 (25)



$$A(x) = \pi(1 - \sin x)^2$$

$$\Delta V = \pi(1 - \sin x)^2 \Delta x$$

$$V = \lim_{\Delta x \rightarrow 0} \sum_{x=0}^{\pi/2} \pi(1 - \sin x)^2 \Delta x$$

$$= \pi \int_0^{\pi/2} (1 - 2\sin x + \sin^2 x) dx$$

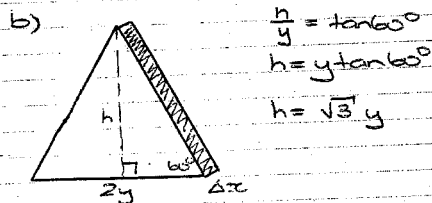
$$= \pi \int_0^{\pi/2} (1 - 2\sin x + \frac{1}{2} - \frac{1}{2}\cos 2x) dx$$

$$= \pi \left[\frac{3}{2}x + 2\cos x - \frac{1}{4}\sin 2x \right]_0^{\pi/2}$$

$$= \pi \left(\frac{3\pi}{4} + 0 - 0 - 0 - 2 + 0 \right)$$

$$= \pi \left(\frac{3\pi}{4} - 2 \right)$$

$$= \frac{3\pi^2 - 8\pi}{4} \text{ units}^3 \quad (5)$$



$$A(x) = \frac{1}{2}(2y)(\sqrt{3}y)$$

$$= \sqrt{3}y^2$$

$$= \sqrt{3}(4 - x^2)$$

$$\Delta V = \sqrt{3}(4 - x^2) \Delta x$$

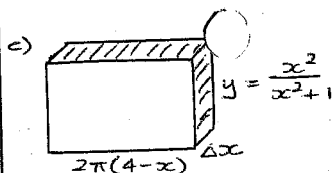
$$V = \lim_{\Delta x \rightarrow 0} \sum_{x=2}^0 \sqrt{3}(4 - x^2) \Delta x$$

$$= 2\sqrt{3} \int_0^2 (4 - x^2) dx$$

$$= 2\sqrt{3} \left[4x - \frac{1}{3}x^3 \right]_0^2$$

$$= 2\sqrt{3} \left(8 - \frac{8}{3} \right)$$

$$= \frac{32\sqrt{3}}{3} \text{ units}^3 \quad (5)$$



$$A(x) = 2\pi(4-x) \left(\frac{x^2}{x^2+1} \right)$$

$$= 2\pi(4-x) \left(1 - \frac{1}{x^2+1} \right) \quad (3)$$

$$\Delta V = 2\pi(4-x) \left(1 - \frac{1}{x^2+1} \right) \Delta x$$

(ii)

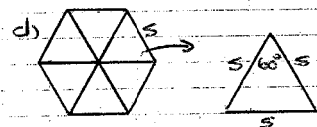
$$V = \lim_{\Delta x \rightarrow 0} \sum_{x=0}^2 2\pi(4-x) \left(1 - \frac{1}{x^2+1} \right) \Delta x$$

$$V = 2\pi \int_0^2 \left[4 - x - \frac{4}{x^2+1} + \frac{x}{x^2+1} \right] dx$$

$$= 2\pi \left[4x - \frac{1}{2}x^2 - 4\tan^{-1}x + \frac{1}{2}\log(x^2+1) \right]_0^2$$

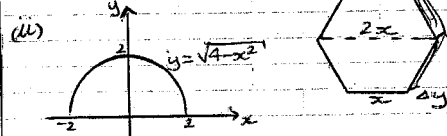
$$= 2\pi(8 - 2 - 4\tan^{-1}2 + \frac{1}{2}\log 5 - 0)$$

$$= 2\pi(6 - 4\tan^{-1}2 + \frac{1}{2}\log 5) \text{ units}^3 \quad (3)$$



$$A = 6 \times \frac{1}{2}(s)(s) \sin 60^\circ$$

$$= \frac{3\sqrt{3}s^2}{2}$$



$$A(y) = \frac{3\sqrt{3}}{2}x^2$$

$$= \frac{3\sqrt{3}}{2}(4 - y^2)$$

$$\Delta V = \frac{3\sqrt{3}}{2}(4 - y^2) \Delta y$$

$$V = \lim_{\Delta y \rightarrow 0} \sum_{y=0}^2 \frac{3\sqrt{3}}{2}(4 - y^2) \Delta y$$

$$= \frac{3\sqrt{3}}{2} \int_0^2 (4 - y^2) dy$$

$$= \frac{3\sqrt{3}}{2} \left[4y - \frac{1}{3}y^3 \right]_0^2$$

$$= \frac{3\sqrt{3}}{2} \left(8 - \frac{8}{3} \right)$$

$$= 8\sqrt{3} \text{ m}^3 \quad (5)$$